

# Configuration and Parameters of the IEEE 14-Bus Test System and the Neural Network Surrogate

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The test system considered in this note is based on the IEEE 14-bus system in MATPOWER. The line and bus parameters follow the standard case14 data, while additional wind-power settings, time-varying load profiles, and green-ratio tracing constraints are introduced for the dispatch and surrogate-model studies. The system base power is 100 MVA. The network contains 14 buses, 19 branches, 5 thermal generating units, and 1 wind generator. The wind generator is connected to Bus 3 and its installed capacity is 100 MW.

We consider a 24-hour daily dispatch problem with the time interval  $\Delta t = 1$  h.

## 1 Generation and Wind

The thermal generator locations, generation limits, and quadratic cost coefficients are listed in Table 1. The generation cost is modeled as

$$C_{\text{gen}} = \sum_{t=1}^T \Delta t \sum_{i=1}^5 (c_{2,i} P_{i,t}^2 + c_{1,i} P_{i,t} + c_{0,i}), \quad (1)$$

where  $P_{i,t}$  is the power output of thermal unit  $i$  at time period  $t$ . In the dispatch model, the quadratic cost is further approximated by a 4-segment piecewise linear representation so that the exact tracing model and the neural-network-embedded model use the same objective and the same generator cost parameters.

The minimum active output of each thermal unit is set to 20% of its maximum output, i.e.,

$$P_i^{\min} = 0.2 P_i^{\max}. \quad (2)$$

The ramp-up and ramp-down limits are both set to  $0.6 P_i^{\max}$  per hour, and the minimum up and down times are both set to 3 h. The startup cost is set to 3000 for each thermal unit.

Table 1: Thermal generator parameters of the IEEE 14-bus test system.

| Unit | Bus | $P^{\max}$ (MW) | $P^{\min}$ (MW) | $c_2$  | $c_1$ | $c_0$ |
|------|-----|-----------------|-----------------|--------|-------|-------|
| TU1  | 1   | 332.4           | 66.48           | 0.0430 | 19    | 0     |
| TU2  | 2   | 140.0           | 28.00           | 0.2500 | 20    | 0     |
| TU3  | 3   | 100.0           | 20.00           | 0.0100 | 40    | 0     |
| TU4  | 6   | 100.0           | 20.00           | 0.0100 | 41    | 0     |
| TU5  | 8   | 100.0           | 20.00           | 0.0100 | 42    | 0     |

The wind-generator configuration is

$$\text{WF} = \{(3, 100 \text{ MW})\}. \quad (3)$$

The day-ahead wind coefficients used in the case study are given by

$$\begin{aligned} \alpha_{\text{wind}} = [ & 0.6602, 0.6830, 0.7083, 0.6925, 0.6845, 0.7142, 0.7235, 0.5415, \\ & 0.7793, 0.7860, 0.7215, 0.6585, 0.6053, 0.6038, 0.6068, 0.9050, \\ & 0.8218, 0.8220, 0.6870, 0.6559, 0.6720, 0.6934, 0.7058, 0.6964]. \end{aligned} \quad (4)$$

These coefficients are derived from the original wind prediction profile and then adjusted by multiplying the values in periods 8–15 by 3 and the values in periods 19–24 by 1.5.

## 2 Load

The base active loads at the 14 buses are given by

$$P_{\text{base}}^L = [0, 21.7, 94.2, 47.8, 7.6, 11.2, 0, 0, 29.5, 9.0, 3.5, 6.1, 13.5, 14.9] \text{ MW}. \quad (5)$$

The corresponding total base active load is 259.0 MW. The day-ahead load coefficients are given by

$$\begin{aligned} \alpha_{\text{load}} = [ & 0.7389, 0.7187, 0.7035, 0.6983, 0.7040, 0.7251, 0.7764, 0.8461, \\ & 0.8951, 0.9094, 0.9078, 0.9071, 0.8923, 0.8877, 0.8840, 0.8849, \\ & 0.8941, 0.9345, 0.9899, 0.9841, 0.9691, 0.9436, 0.9094, 0.8674]. \end{aligned} \quad (6)$$

The active load at each bus and each hour is obtained by scaling the corresponding base load with the above time-varying coefficient. In the training-data generation process for the neural network, the total load of each sample is set equal to the total thermal generation plus the total wind generation, and then allocated to each load bus in proportion to the base-load distribution.

## 3 Neural Network Surrogate

A feedforward neural network is used to approximate the green-power ratio at the key load buses. The network input contains only the power quantities directly available from the dispatch result, namely,

$$\mathbf{x} = [P_{\text{wind}}^{\Sigma}, P_{\text{TU}1}, P_{\text{TU}2}, P_{\text{TU}3}, P_{\text{TU}4}, P_{\text{TU}5}]^{\top}, \quad (7)$$

where  $P_{\text{wind}}^{\Sigma}$  is the total wind generation and  $P_{\text{TU}i}$  is the output of thermal unit  $i$ . The network output is

$$\hat{\mathbf{y}} = [\hat{w}_3, \hat{w}_4]^{\top}, \quad (8)$$

where  $\hat{w}_3$  and  $\hat{w}_4$  denote the estimated green-power ratios at Bus 3 and Bus 4, respectively. Therefore, the surrogate used in this study is a 6-input, 2-output model.

The training labels are generated by the exact proportional-sharing model with the same physical parameters as those used in the dispatch model. The full dataset stores the green ratios for all 14 buses, while only the outputs at Bus 3 and Bus 4 are extracted for neural network training. Very small target values below 0.02 are clipped to zero to suppress numerical noise.

The dataset generation settings are as follows. All unit-commitment patterns with 1, 2, and 3

online thermal units are enumerated. The wind level is sampled at 20 uniformly spaced points from 0.1 to 1.0 of the installed wind capacity. For each online thermal unit, the feasible active output range is discretized into 40 segments. If the number of joint output allocations for a multi-unit commitment is too large, at most 50 operating points are randomly retained. Using this procedure, a total of 16299 valid samples are generated. The dataset is split into 70% for training, 15% for validation, and 15% for testing, corresponding to 11409, 2444, and 2446 samples, respectively.

The neural network hyperparameters are listed in Table 2. The model uses two hidden layers with 24 and 12 neurons, respectively, and ReLU activation functions. The parameters are optimized by Adam with an initial learning rate of  $5 \times 10^{-4}$ . The loss function is the Huber loss. Early stopping is used with a patience of 200 epochs, and a quadratic calibration polynomial is fitted on the validation set after training.

Table 2: Neural network hyperparameters.

| Parameter                        | Value                |
|----------------------------------|----------------------|
| Input dimension                  | 6                    |
| Output dimension                 | 2                    |
| Output buses                     | Bus 3, Bus 4         |
| Hidden-layer structure           | [24, 12]             |
| Activation function              | ReLU                 |
| Optimizer                        | Adam                 |
| Initial learning rate            | $5 \times 10^{-4}$   |
| Maximum training epochs          | 1000                 |
| Batch size                       | 128                  |
| $L_2$ regularization coefficient | $5 \times 10^{-4}$   |
| Loss function                    | Huber                |
| Early stopping patience          | 200                  |
| Data split                       | 70% / 15% / 15%      |
| Post-training calibration        | Quadratic polynomial |
| Calibration degree               | 2                    |

For mixed-integer embedding into the dispatch model, the trained ReLU neural network is reformulated as linear inequalities with binary activation variables. Since the two hidden layers contain 24 and 12 neurons, respectively, the embedded neural network introduces 36 binary variables associated with the ReLU activation states.